Lab 05 - Combinatorial Logic

In this lab, you've learned real world applications of digital logic, as well as how to assemble your own Verilog modules. In addition, you've learned how the constraints file maps your inputs and outputs to real pins on the FPGA.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab 5 Combinatorial Logic

In this lab, we determined the maxterm and minterm of 2 truth tables and edited our circuit A and B codes. Next, we edited the top file to have sw[6, 0] and led[0,1], with a wire connecting the output of A to the A input of B. The final step before Vivado was to edit the constraint file to use the sw and led that we used. The last step was to create a project in Vivado and to implement all of our files onto the project, run the project, and program the BASYS3 board.

Lab Questions

1 - Explain the role of the Top Level file.
The role of the top level file is to assign the switches to both circuits. Then we use the top file to add the wire so that both of the circuits are combined to one circuit so that we can run it on the Basys.

2 - Explain the function of the Constraints file.
It lets you select the number of switches and led you would like to use in the project. For this lab it was switch 0-6 and the led was 0 and 1. So only those were uncommented to activate those specific properties.

3 - Was the selection of Minterm and Maxterm correct for each circuit? What would you have chosen?

Yes, the selection for both of the minterm and maxterm for each of the circuits was correct. I would have chosen to use minterm because the idea of it works better for me.